3.3 Solving Quadratic Equations

Name _____ Period ____ Date _____

Algebra II 3.3 · Student sheet

I can…

- solve quadratic equations using square roots.
- solve quadratic equations by factoring and by graphing.
- choose a method and use zeros to solve real-life problems.

Vocabulary · fill in the missing word

Quadratic equation — an equation that can be _____ $ax^2 + bx + c = 0$, $a \neq 0$

Root of an equation — a _____

Zero of a function — an _____ where $f(x) = 0$

Square roots — if $x^2 =$ _____, then $x = \pm\sqrt{d}$

Part 1 · Three ways

Solve $x^2 - 2x - 8 = 0$ three ways.

Method	Work	Solutions
Graphing (Desmos)		
Factoring		
Table of values		

Which way would you use next time? Why? _____

Part 2 · Notes: fill in the blanks

If $x^2 = d$ $(d > 0)$, then $x =$ _____.

Before factoring, the equation must equal _____.

On a graph, the solutions are the _____.

Part 3 · Examples (work them with the class)

Example 1

Solve each equation.

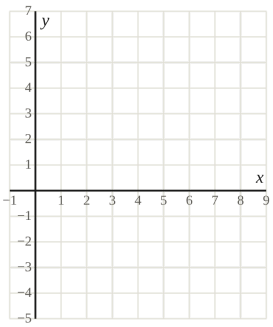
a. $2x^2 - 50 = 0$ b. $(x - 3)^2 = 7$

Example 2

Solve $x^2 + 3x = 18$.

Example 3

Find the zeros of $f(x) = x^2 - 8x + 12$. Check with the graph.



Example 4

A store sells 100 hats a week at $20 each. For each $1 increase in price, it sells 4 fewer hats. What price maximizes the weekly revenue? What is the maximum revenue?

Part 4 · Partner practice

Partner 1

Solve $3x^2 = 48$.

Partner 2

Find the zeros of $g(x) = x^2 + x - 12$.